

RQ1: Wildfire Burned Area and Annual Mean AQI in California

Abrão Soares Pereira

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Research Question 1

Is total wildfire burned area associated with annual mean AQI at the county level? We hypothesize that larger burned areas will be positively associated with higher mean AQI, after controlling for social vulnerability and year.

1. Load Packages

```
library(here)
library(tidyverse)
library(nlme)
library(car)
library(gtsummary)
```

2. Load and Inspect Data

```
rq1_data <- read.csv("~/Documents/ENVIRON 710/ENV710_Group12/final_dataframe.csv")
```

3. Organize data

```
cleaned_data_rq1 <- rq1_data %>%
  rename(
    county = COUNTY,
    year = Year,
    mean_aqi = AQI,
    acres = GIS_ACRES,
    svi = RPL_THEMES
  ) %>%
  mutate(
    # Log10 transform burned area; zero-fire county-years set to log10(1) = 0
    acres = ifelse(is.na(acres) | acres == 0, 1, acres),
    log_acres = log10(acres),
    year = as.integer(year),
    county = as.factor(county)
```

```

) %>%
  filter(!is.na(mean_aqi), !is.na(log_acres))

cat("Rows:", nrow(cleaned_data_rq1), "\n")

## Rows: 188

cat("Counties:", n_distinct(cleaned_data_rq1$county), "\n")

## Counties: 53

cat("Years:", paste(sort(unique(cleaned_data_rq1$year)), collapse = ", "), "\n")

## Years: 2016, 2018, 2020, 2022

# Check missing data
cleaned_data_rq1 %>%
  select(mean_aqi, log_acres, svi, year) %>%
  summarise(across(everything(), ~sum(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "n_missing") %>%
  print()

## # A tibble: 4 x 2
##   variable n_missing
##   <chr>      <int>
## 1 mean_aqi      0
## 2 log_acres      0
## 3 svi           0
## 4 year          0

# Summarize data by year for the statewide trend plot
annual <- cleaned_data_rq1 %>%
  group_by(year) %>%
  summarise(
    mean_aqi_state = mean(mean_aqi, na.rm = TRUE),
    total_acres     = sum(acres, na.rm = TRUE),
    .groups = "drop"
  )

```

4. Exploratory Data Analysis

4.1 Distribution of Key Variables

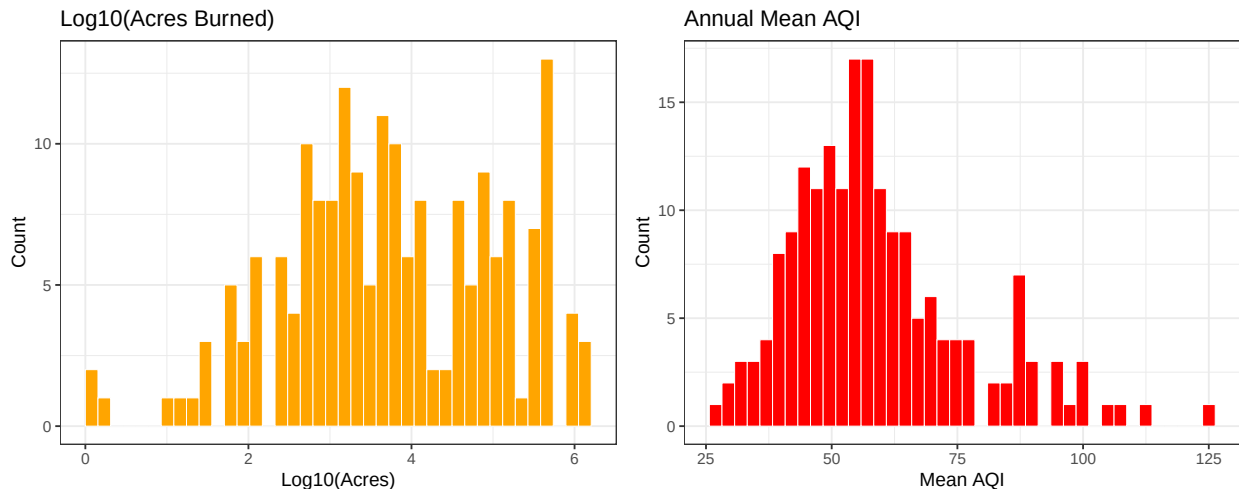
```

# Graphs and plots
graph_01 <- ggplot(cleaned_data_rq1, aes(x = log_acres)) +
  geom_histogram(bins = 40, fill = "orange", color = "white", linewidth = 0.2) +
  labs(title = "Log10(Acres Burned)", x = "Log10(Acres)", y = "Count") +
  theme_bw()

```

```
graph_02 <- ggplot(cleaned_data_rq1, aes(x = mean_aqi)) +
  geom_histogram(bins = 40, fill = "red", color = "white", linewidth = 0.2) +
  labs(title = "Annual Mean AQI", x = "Mean AQI", y = "Count") +
  theme_bw()

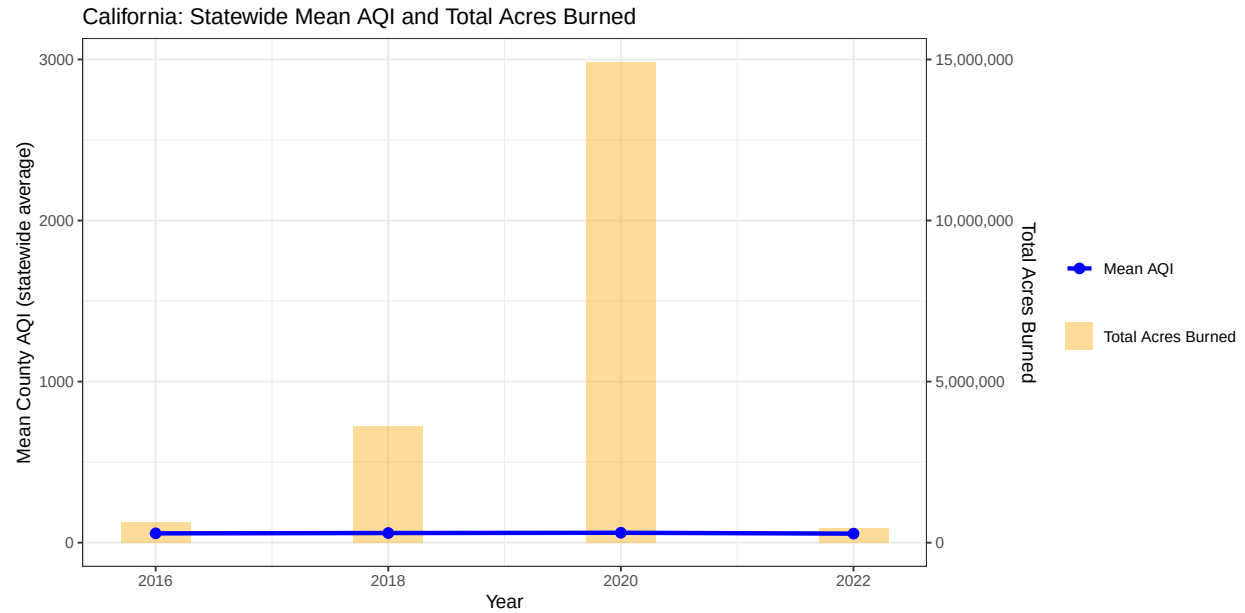
gridExtra::grid.arrange(graph_01, graph_02, ncol = 2)
```



4.2 Statewide Trend Over Time

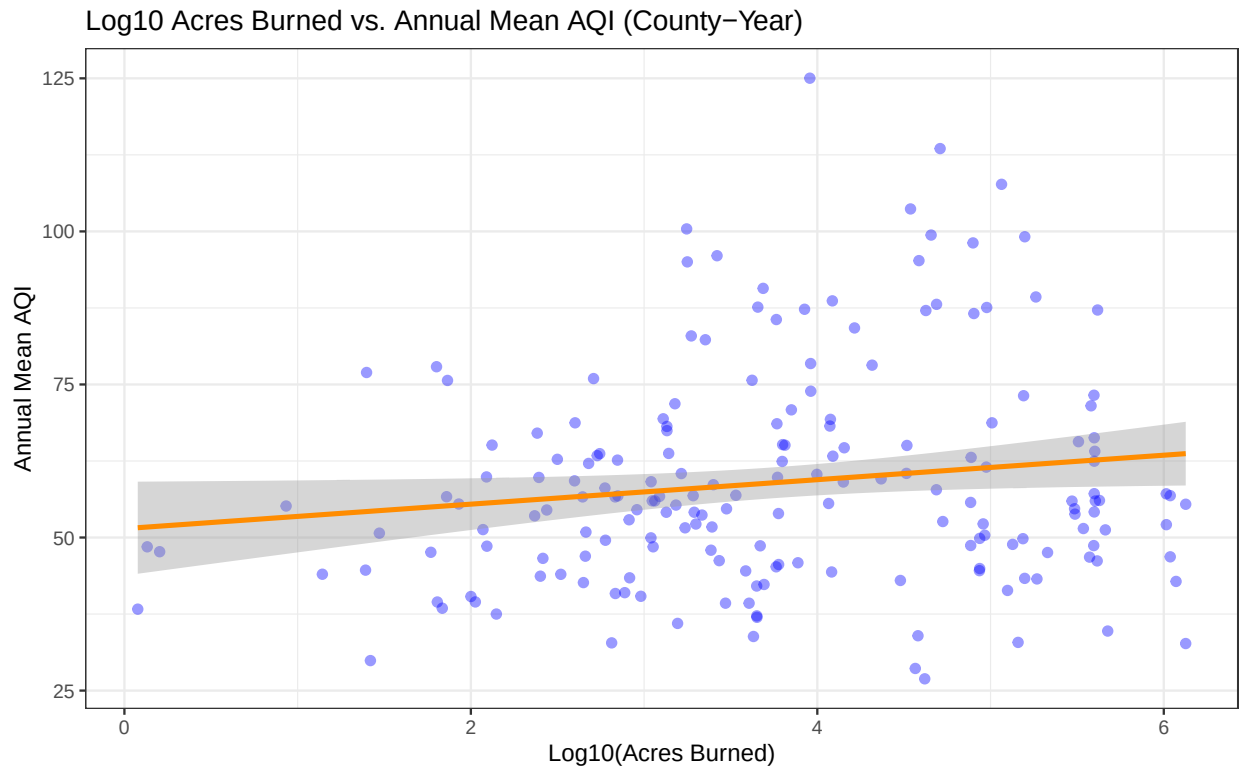
```
# Dual axis plot

ggplot(annual, aes(x = year)) +
  geom_col(aes(y = total_acres / 5000, fill = "Total Acres Burned"),
    alpha = 0.4, width = 0.6) +
  geom_line(aes(y = mean_aqi_state, color = "Mean AQI"),
    linewidth = 1.2) +
  geom_point(aes(y = mean_aqi_state, color = "Mean AQI"),
    size = 2.5) +
  scale_fill_manual(values = c("Total Acres Burned" = "orange")) +
  scale_color_manual(values = c("Mean AQI" = "blue")) +
  scale_y_continuous(
    name = "Mean County AQI (statewide average)",
    sec.axis = sec_axis(~. * 5000, name = "Total Acres Burned",
      labels = scales::comma)
  ) +
  labs(
    title = "California: Statewide Mean AQI and Total Acres Burned",
    x = "Year",
    fill = NULL,
    color = NULL
  ) +
  theme_bw()
```



4.3 Scatter: Log Acres Burned vs. Mean AQI

```
# Scatter plot for burned areas and mean AQI
ggplot(cleaned_data_rq1, aes(x = log_acres, y = mean_aqi)) +
  geom_point(alpha = 0.4, color = "blue", size = 1.8) +
  geom_smooth(method = "lm", color = "darkorange", linewidth = 1.1, se = TRUE) +
  labs(
    title = "Log10 Acres Burned vs. Annual Mean AQI (County-Year)",
    x     = "Log10(Acres Burned)",
    y     = "Annual Mean AQI"
  ) +
  theme_bw()
```



5. Multicollinearity Check (VIF)

```
# Check for multicollinearity
vif_model <- lm(mean_aqi ~ log_acres + svi + year, data = cleaned_data_rq1)

vif_results <- car::vif(vif_model)
print(round(vif_results, 3))
```

```
## log_acres      svi      year
##      1.029      1.025      1.005
```

```
flagged <- vif_results[vif_results > 5]
if (length(flagged) == 0) {
  cat("No variables exceed VIF > 5 - all predictors acceptable.\n")
} else {
  print(flagged)
}
```

```
## No variables exceed VIF > 5 - all predictors acceptable.
```

6. Linear Mixed-Effects Models

6.1 Fit Standard Linear Regression (for AIC comparison)

```
# compare GLS with LME
cali_gls <- gls(mean_aqi ~ log_acres + year, data = cleaned_data_rq1, method = "ML")
```

6.2 Base Model: Burned Area + Year

```
# Simple mixed model
model_base <- lme(
  mean_aqi ~ log_acres + year,
  random = ~1 | county,
  data = cleaned_data_rq1,
  method = "ML"
)

summary(model_base)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: cleaned_data_rq1
##       AIC       BIC    logLik
##  1443.97 1460.152 -716.985
##
## Random effects:
## Formula: ~1 | county
##      (Intercept) Residual
## StdDev:      15.84172  7.356035
##
## Fixed effects: mean_aqi ~ log_acres + year
##              Value Std.Error DF   t-value p-value
## (Intercept)  594.8121  524.8526 133   1.133294  0.2591
## log_acres      1.4173    0.4994 133   2.837815  0.0053
## year          -0.2683    0.2601 133  -1.031468  0.3042
## Correlation:
##      (Intr) lg_crs
## log_acres  0.112
## year      -1.000 -0.116
##
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -6.27223824 -0.34102289 -0.03685504  0.34767580  4.40116366
##
## Number of Observations: 188
## Number of Groups: 53
```

6.3 Extended Model: Add SVI as Covariate

```
# Extended model with SVI
model_svi <- lme(
  mean_aqi ~ log_acres + svi + year,
  random = ~1 | county,
  data = cleaned_data_rq1,
```

```

method = "ML"
)

summary(model_svi)

## Linear mixed-effects model fit by maximum likelihood
##   Data: cleaned_data_rq1
##       AIC      BIC    logLik
## 1440.666 1460.084 -714.3329
##
## Random effects:
##   Formula: ~1 | county
##       (Intercept) Residual
## StdDev:      14.90643  7.375388
##
## Fixed effects: mean_aqi ~ log_acres + svi + year
##              Value Std.Error   DF   t-value p-value
## (Intercept) 634.8390  527.7203 132   1.202984  0.2311
## log_acres      1.4486   0.5013 132   2.889853  0.0045
## svi           14.3985   6.1178 132   2.353550  0.0201
## year          -0.2918   0.2615 132  -1.115476  0.2667
## Correlation:
##      (Intr) lg_crs svi
## log_acres  0.113
## svi        0.035  0.025
## year      -1.000 -0.116 -0.041
##
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -6.0658872 -0.3144252 -0.0313626  0.3544790  4.4701578
##
## Number of Observations: 188
## Number of Groups: 53

```

6.4 Model Comparison (AIC)

```

# compare models AIC
AIC(cali_gls, model_base, model_svi)

```

```

##           df      AIC
## cali_gls    4 1616.106
## model_base  5 1443.970
## model_svi   6 1440.666

```

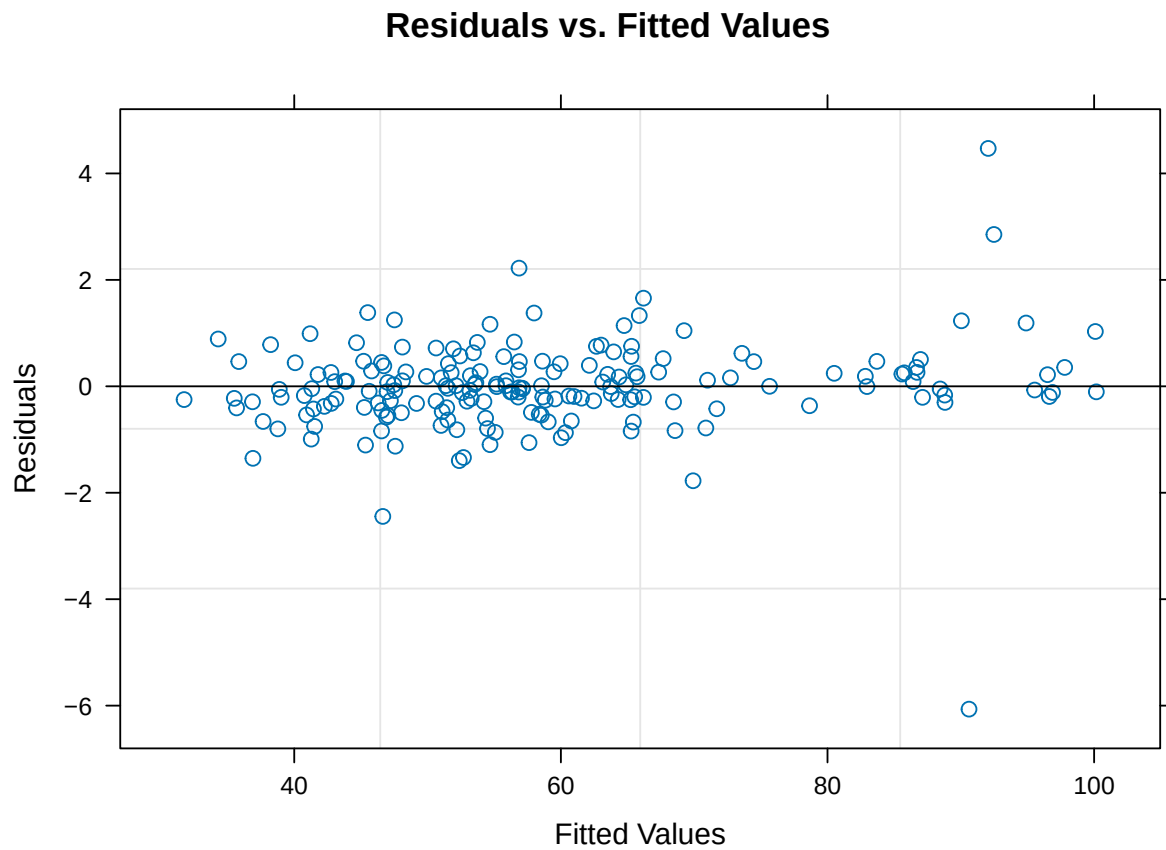
7. Model Diagnostics

```

# Best model fitting
best_model <- model_svi

```

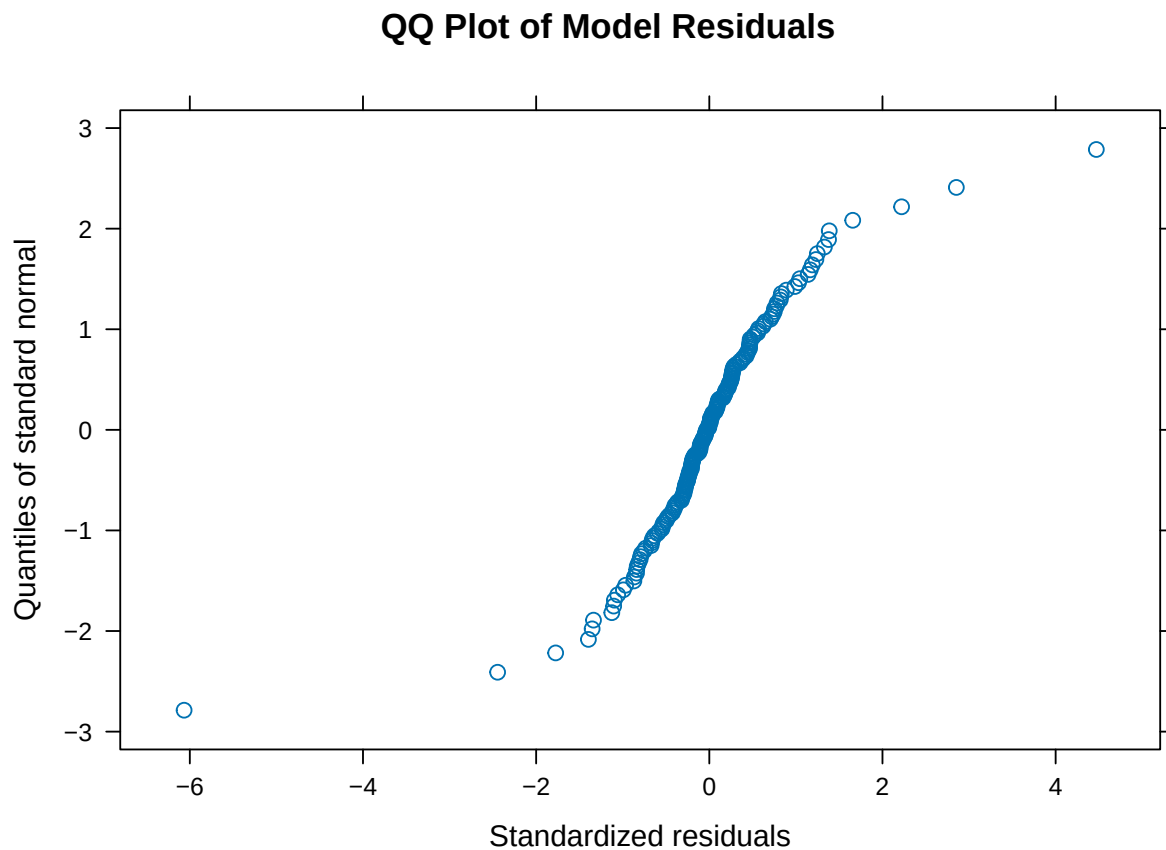
```
# Residuals vs. Fitted (homoscedasticity)  
plot(best_model,  
      xlab = "Fitted Values",  
      ylab = "Residuals",  
      main = "Residuals vs. Fitted Values")
```



```
# QQ Plot (normality of residuals)  
qqnorm(best_model, main = "QQ Plot of Model Residuals")
```


Characteristic	Beta	95% CI	p-value
(Intercept)	635	-398, 1,668	0.2
log_acres	1.4	0.47, 2.4	0.005
svi	14	2.4, 26	0.020
year	-0.29	-0.80, 0.22	0.3
county.SD (Intercept)	15		
Residual.SD (Observations)	7.4		

Abbreviation: CI = Confidence Interval



8. Results Table

```
# Create a table
tbl_regression(best_model)
```

9. Results Visualization

```

# visualize model result
predict_data_rq1 <- data.frame(
  log_acres = seq(min(cleaned_data_rq1$log_acres), max(cleaned_data_rq1$log_acres), length.out = 200),
  svi       = mean(cleaned_data_rq1$svi, na.rm = TRUE),
  year      = median(cleaned_data_rq1$year)
)

predict_data_rq1$predicted_aqi <- predict(best_model, newdata = predict_data_rq1, level = 0)

ggplot() +
  geom_point(data = cleaned_data_rq1, aes(x = log_acres, y = mean_aqi),
            alpha = 0.3, color = "blue", size = 1.5) +
  geom_line(data = predict_data_rq1, aes(x = log_acres, y = predicted_aqi),
           color = "orange", linewidth = 1.3) +
  labs(
    title   = "Model-Predicted Mean AQI by Wildfire Burned Area",
    subtitle = "Population-level prediction (average SVI county, median year)",
    x       = "Log10(Acres Burned)",
    y       = "Annual Mean AQI"
  ) +
  theme_bw()

```

